Laptop Setup Guide

From Screen to Scene: Building VR Experiments with Python

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Please complete all steps **before** the workshop. The entire process takes about 15 minutes. If you run into problems, check the Troubleshooting section at the end or contact the workshop organizers.

What you need

- A laptop (macOS or Windows) with administrator access
- An internet connection (for downloading software)
- About 1 GB of free disk space

Step 1: Install Python

You need **Python 3.11** (not 3.12 or newer — the 3D engine we use requires 3.11).

macOS

Option A — Download the installer (recommended)

1. Go to https://www.python.org/downloads/
2. Click the big yellow "Download Python 3.1x.x" button
3. Open the downloaded `.pkg` file and follow the installer

Option B — Homebrew (if you already use it)

```
brew install python@3.11
```

Windows

1. Go to https://www.python.org/downloads/
2. Click "Download Python 3.1x.x"
3. Run the installer

4. **Important:** Check the box **"Add Python to PATH"** at the bottom of the first screen before clicking "Install Now"

Verify Python

Open a terminal (macOS: Terminal.app) or Command Prompt (Windows: search for `cmd`), then type:

```
python3 --version        # macOS
python --version         # Windows
```

You should see `Python 3.11.x`. If you see a different major version, `Python 2.x`, or "command not found", revisit the installation steps.

Step 2: Install uv (package manager)

We use **uv** — a fast Python package manager — to set up the project environment.

macOS

Open Terminal and run:

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

Close and reopen Terminal after installation so the `uv` command becomes available.

Windows

Open PowerShell and run:

```
powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1 | iex"
```

Close and reopen PowerShell after installation.

Verify uv

```
uv --version
```

You should see a version number (e.g., `uv 0.7.x`).

Step 3: Download the workshop materials

We will provdee a ZIP file containing all the code and assets for the workshop exercises.

Unzip the downloaded file to a location you can find easily (e.g., Desktop or Documents)

Step 4: Create a virtual environment

Open a terminal and navigate to the project folder:

macOS

```
cd ~/Desktop/vr-tutorial       # adjust path to where you put the folder
uv venv
```

Windows (Command Prompt)

```
cd %USERPROFILE%\Desktop\vr-tutorial
uv venv
```

Windows (PowerShell)

```
cd ~\Desktop\vr-tutorial
uv venv
```

This creates a `.venv` folder inside the project directory.

Step 5: Activate the virtual environment

macOS

```
source .venv/bin/activate
```

Windows (Command Prompt)

```
.venv\Scripts\activate
```

Windows (PowerShell)

```
.venv\Scripts\Activate.ps1
```

Your terminal prompt should now start with `(.venv)`. This means the environment is active.

> **Note:** You need to activate the environment every time you open a new terminal window during the workshop. Keep this command handy.

Step 6: Install dependencies

With the virtual environment activated (you see `(.venv)` in your prompt), run:

```
uv pip install ursina pygame
```

This installs:

- **Ursina** — a 3D game engine we use to build VR scenes
- **pygame** — used for gamepad input in later exercises

The download is about 130 MB. Wait until it finishes.

Step 7: Verify your setup

Run the test script:

macOS

```
python exercises/ex1_hello_ursina/hello_cube.py
```

Windows

```
python exercises\ex1_hello_ursina\hello_cube.py
```

Expected result: A window opens showing an orange cube on a gray background.

Try these mouse controls:

- **Middle-click + drag** — orbit around the cube
- **Scroll wheel** — zoom in and out

Close the window when done (click the X or press Escape).

If you see the cube, you are ready for the workshop!

Quick reference card

Here is a summary of the commands you will use during the workshop:

	macOS	Windows
Terminal	Terminal.app	cmd / PowerShell
Go to project	`cd ~/Desktop/vr-tutorial`	`cd Desktop\vr-tutorial`
Activate env	`source .venv/bin/activate`	`.venv\Scripts\activate`
Run exercise	`python exercises/ex1/hello_cube.py`	`python exercises\ex1\hello_cube.py`
Install pkg	`uv pip install <name>`	`uv pip install <name>`

Troubleshooting

"python" or "python3" not found

- **macOS:** Try `python3` instead of `python`. If neither works, reinstall Python from python.org.
- **Windows:** You likely missed the "Add Python to PATH" checkbox. Rerun the installer, select "Modify", and make sure PATH is checked.

"uv" not found

Close your terminal and reopen it. The `uv` installer updates your PATH, but the change only takes effect in new terminal sessions.

ModuleNotFoundError: No module named 'ursina'

Your virtual environment is not activated. Look for `(.venv)` at the start of your prompt. If it is missing, activate the environment again (Step 5).

The Ursina window opens and closes immediately

- Check the terminal for error messages.
- Make sure you are running the correct script path.
- Update your graphics drivers (Ursina needs OpenGL support).

macOS asks for permissions

macOS may show dialogs about screen recording or accessibility. Click **Allow** — Ursina needs these permissions to open a window and capture mouse input. You can manage these in **System Settings > Privacy & Security**.

Windows Defender or antivirus blocks the script

Some antivirus tools flag Ursina's window creation. Either add the project folder to your antivirus exclusion list, or temporarily disable real-time scanning while running exercises.

Installation fails behind a corporate firewall

If `uv pip install` times out or shows SSL errors, try:

```
uv pip install --trusted-host pypi.org --trusted-host files.pythonhosted.org ursina pygame
```

Gamepad not detected (Exercise 4 only)

1. Plug in the gamepad **before** starting the script.
2. On macOS, check **System Settings > Privacy & Security > Input Monitoring**.
3. Try a different USB port.
4. Gamepad is optional — all exercises work with keyboard and mouse.

Getting help

If none of the above solves your problem, bring your laptop to the workshop. We will have time at the start to debug setup issues together.